StayZen Hotel Management System

Abstract

Efficient management of hotel operations is essential for delivering a seamless guest experience. To address challenges in handling room reservations, event bookings, restaurant services, and customer management, this project proposes StayZen, a comprehensive webbased hotel management system. The platform provides an integrated solution for managing hotel rooms, restaurant reservations, event bookings, guest information, and administrative operations through a centralized dashboard.

The system enables guests to browse available rooms, make reservations, book events, and access restaurant services online. Administrators can efficiently manage room availability, booking schedules, customer records, payments, and service requests. StayZen is designed to streamline hotel operations, reduce manual workload, and enhance customer satisfaction through an intuitive and user-friendly interface.

The application is developed using the Django framework with a responsive Bootstrap-based frontend. The system employs a relational database for secure data storage and efficient management of hotel-related information. Advanced features such as real-time room availability tracking, booking management, event scheduling, and restaurant reservation handling contribute to improved operational efficiency and service quality.

TECHNOLOGY STACK

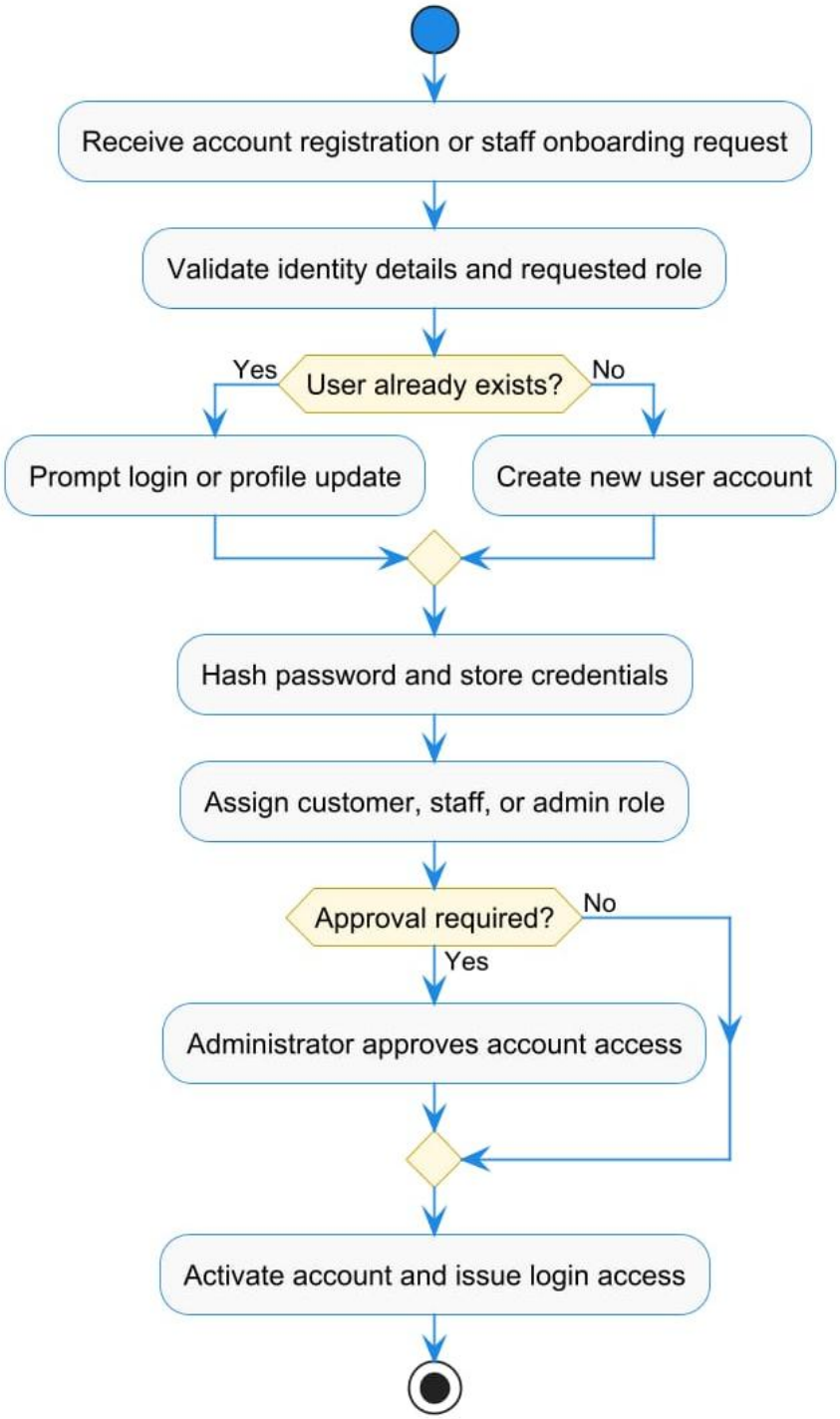
- **Frontend:** HTML, CSS, Bootstrap
- **Backend:** Python (Django)
- **Database:** MySQL
- **Tools & IDE:** VS Code
- **Version Control:** Git & GitHub

Modules Overview

Activity Diagram

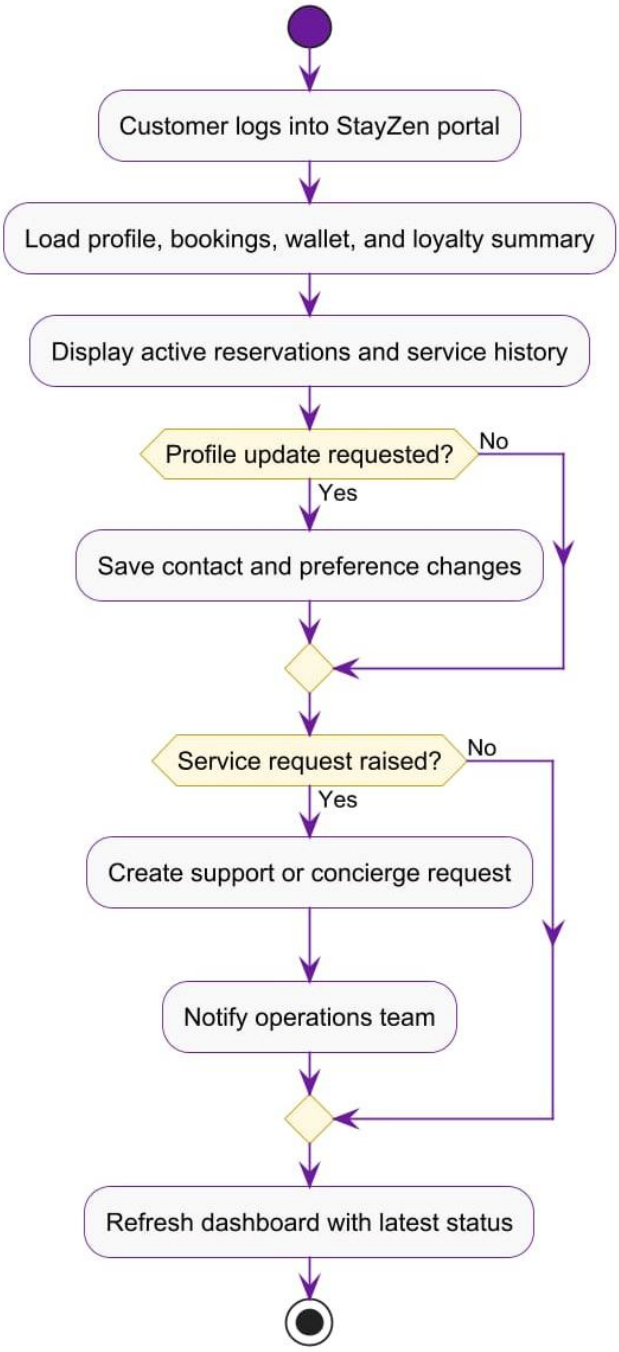
1. User Management Module

Manages user accounts, authentication, authorization, and role-based access control for administrators, staff, and customers.



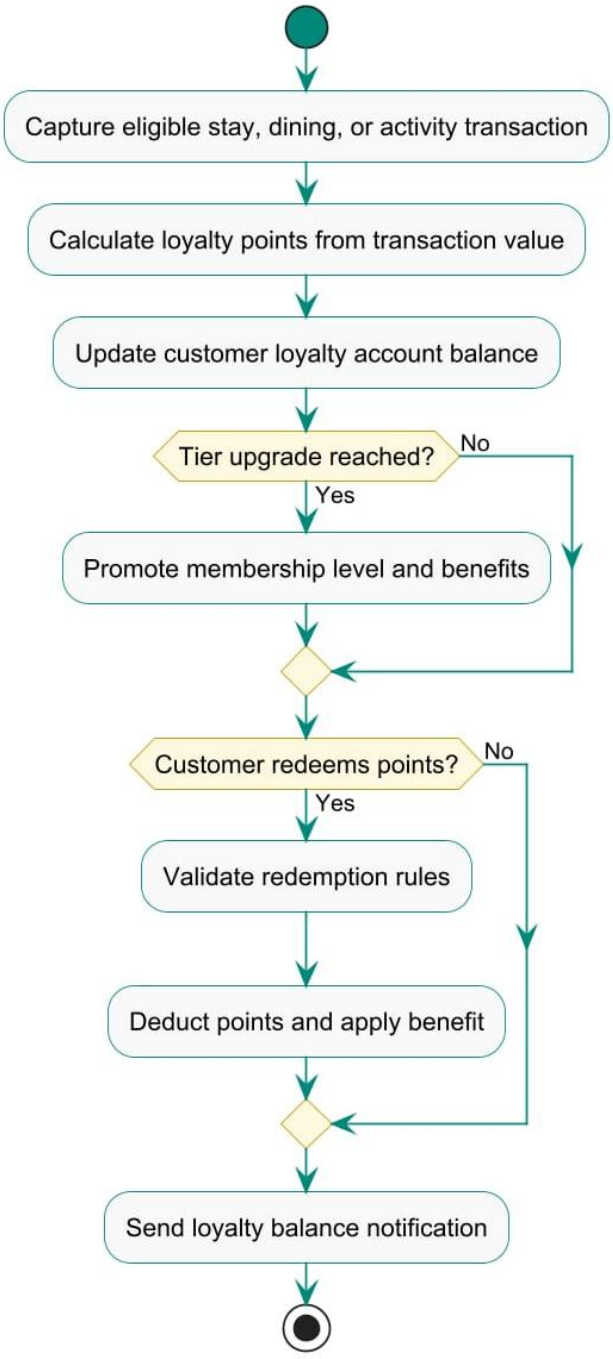
2. Customer Dashboard Module

Provides customers with a personalized interface to manage profiles, bookings, reservations, loyalty points, and account settings.



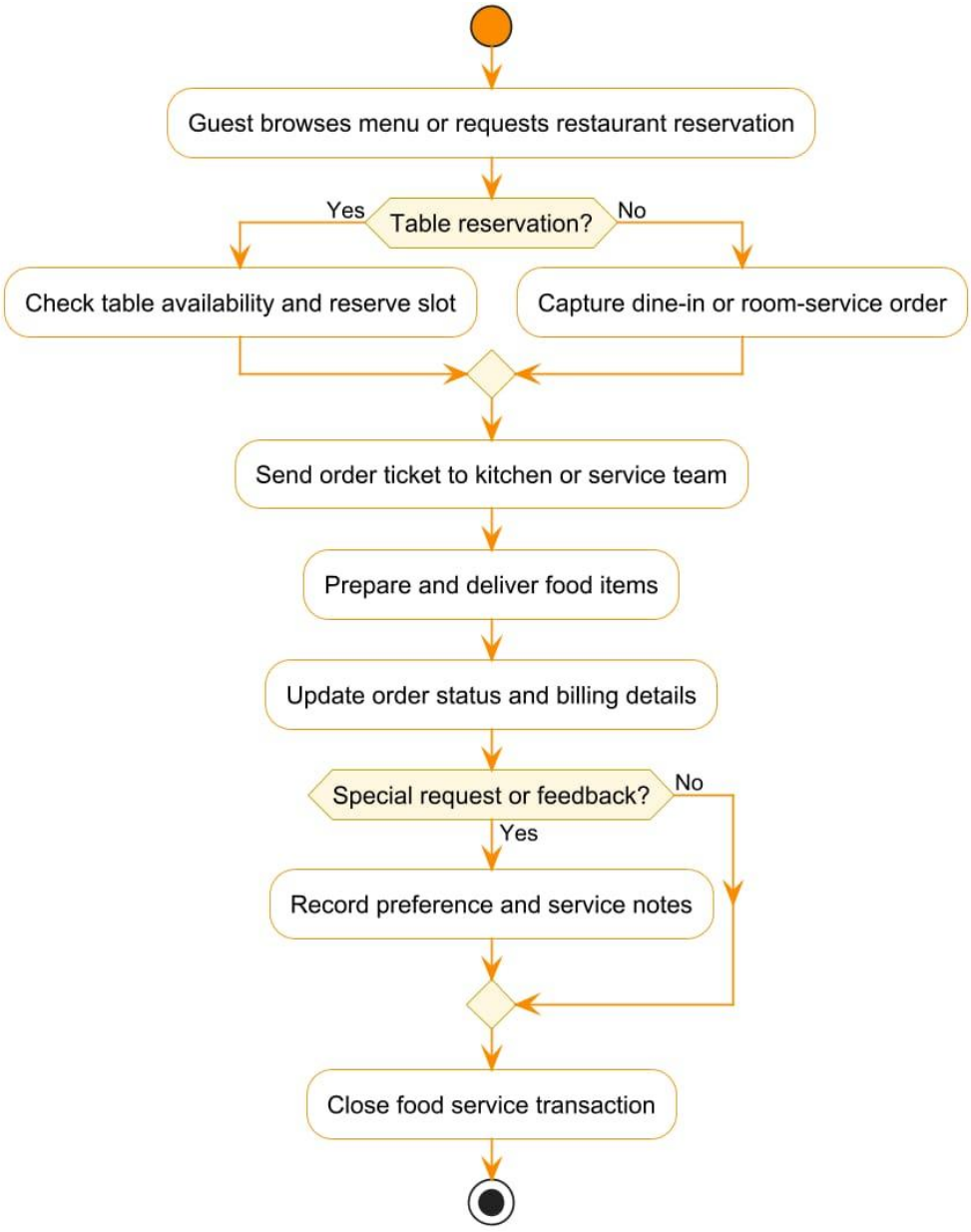
3. Loyalty Management Module

Manages reward points, membership levels, customer benefits, and loyalty program transactions.



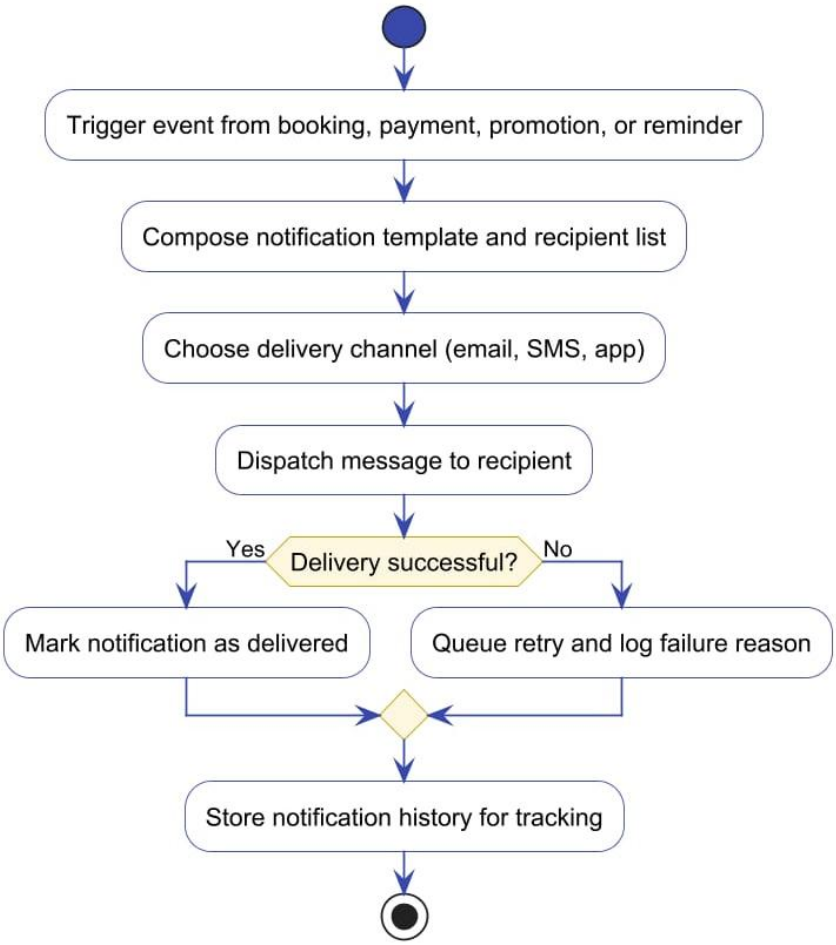
4. Food & Beverage Management Module

Supports restaurant operations, menu management, order processing, and food service tracking.



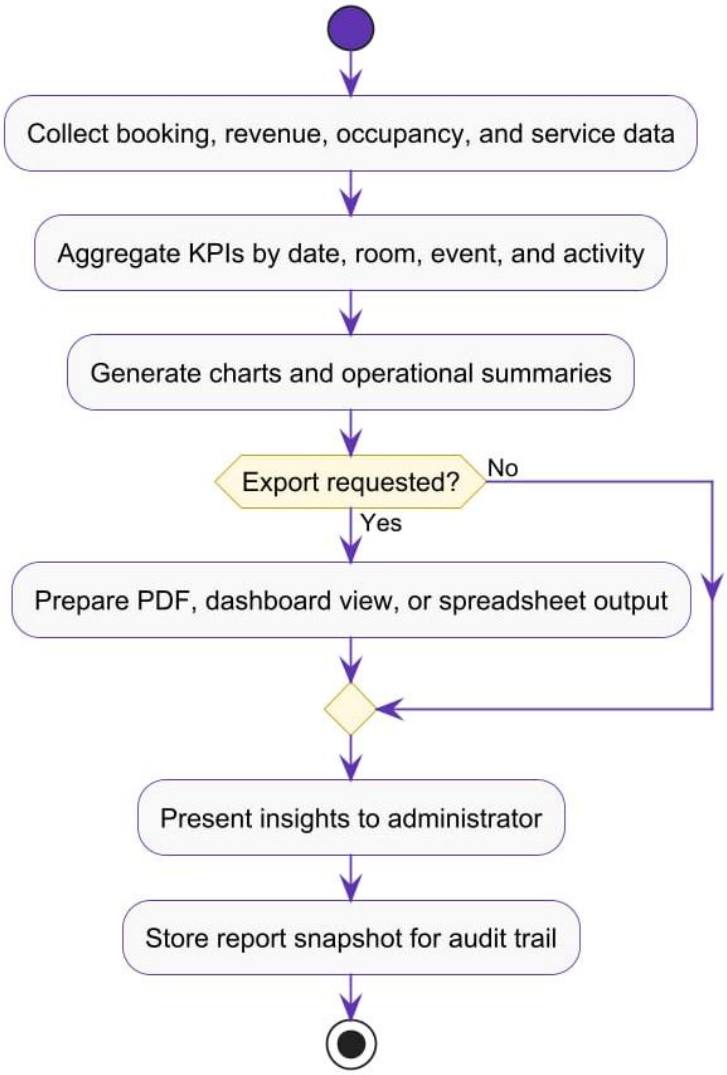
5. Notification Management Module

Sends automated notifications and alerts related to bookings, payments, promotions, and system updates.



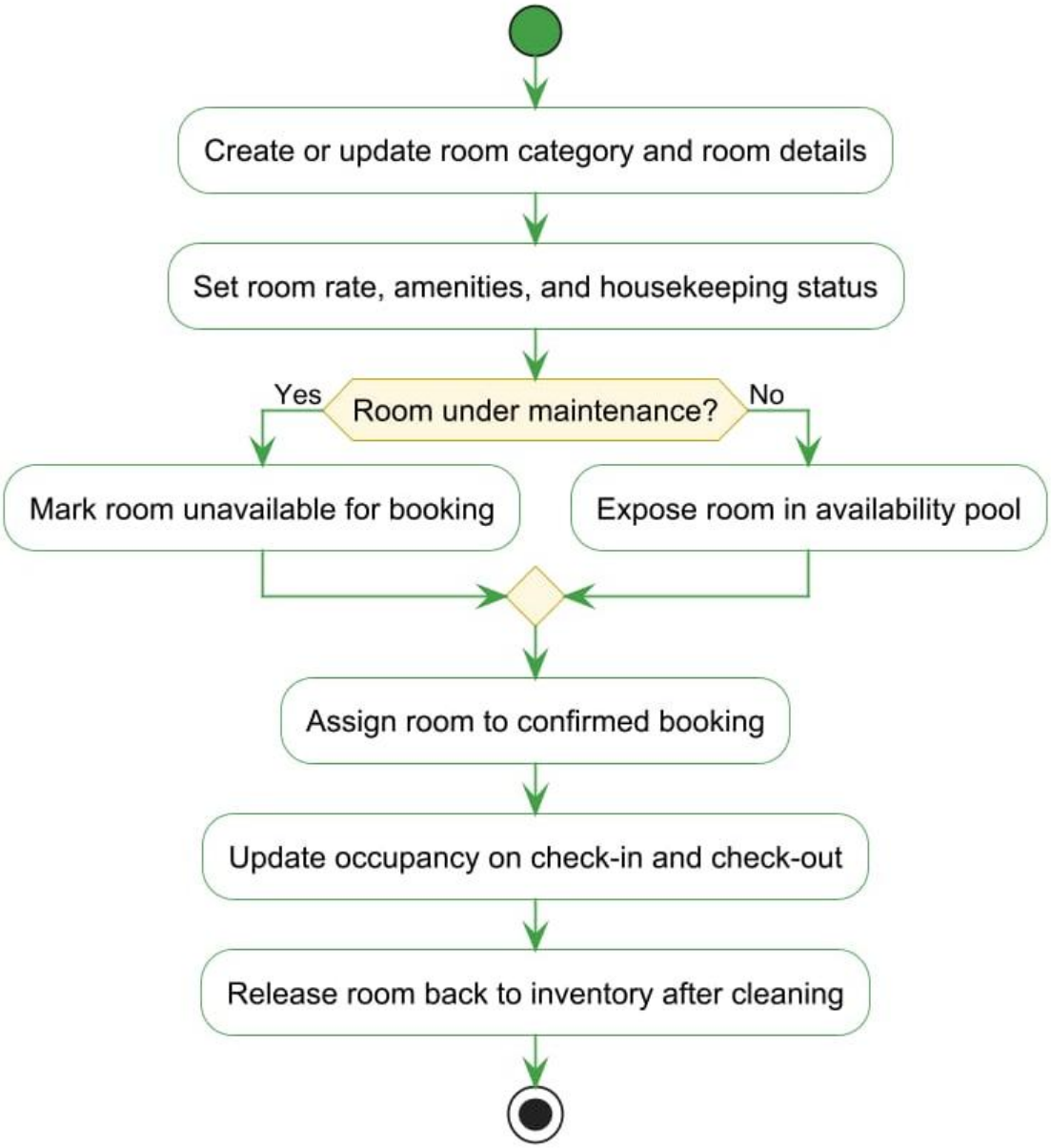
6. Reports & Analytics Module

Generates reports and analytical insights on hotel performance, revenue, bookings, customer trends, and operational activities.



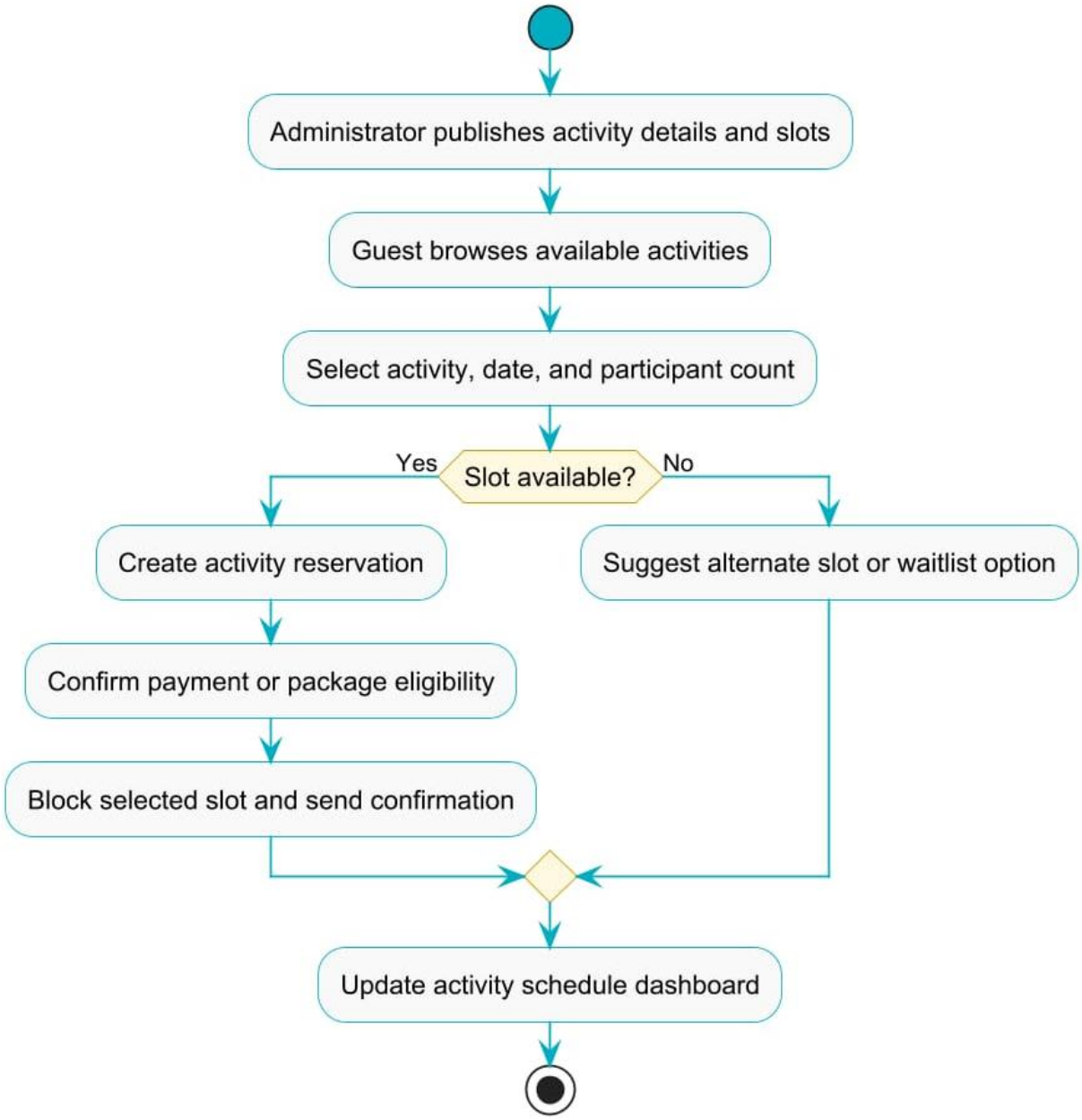
7. Room Management Module

Manages room details, room categories, occupancy status, availability, and room allocation.



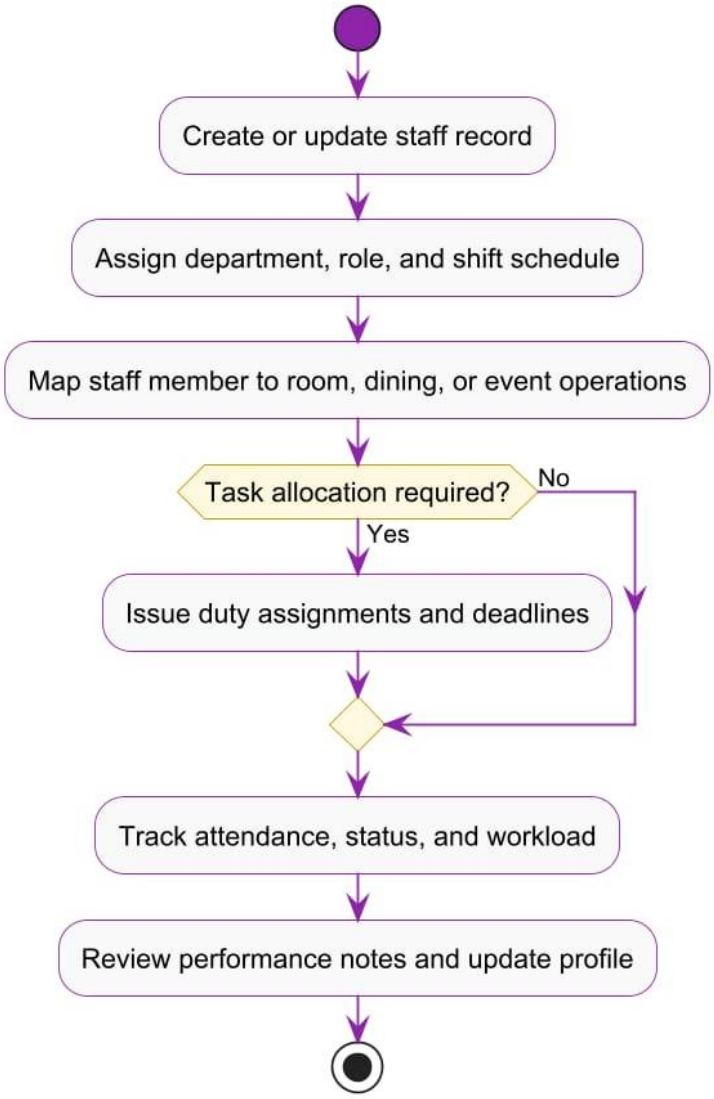
8. Activity Management Module

Allows administrators to manage hotel activities and enables guests to browse and book available activities.



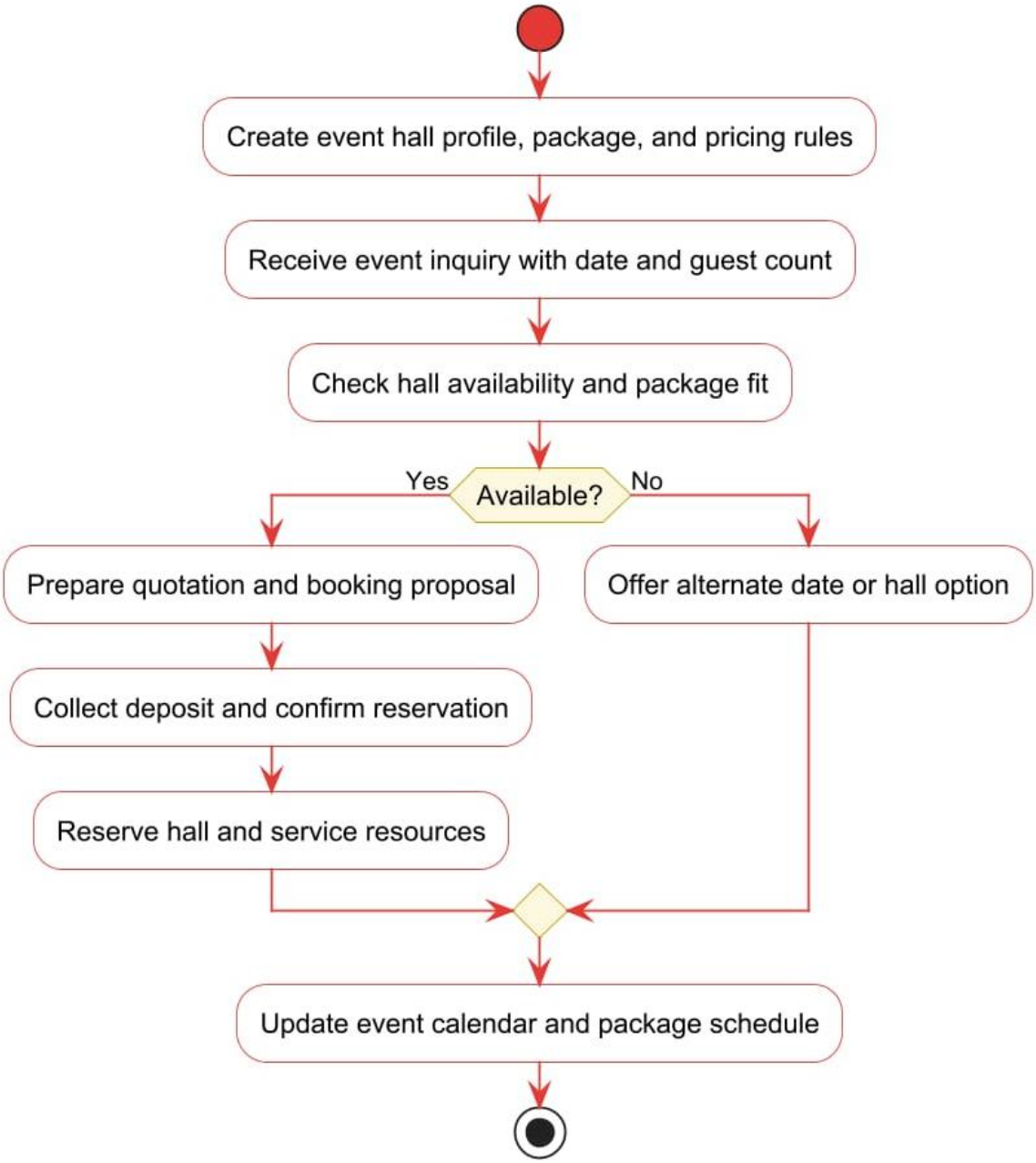
9. Staff Management Module

Maintains employee information, work assignments, roles, schedules, and staff-related operations.



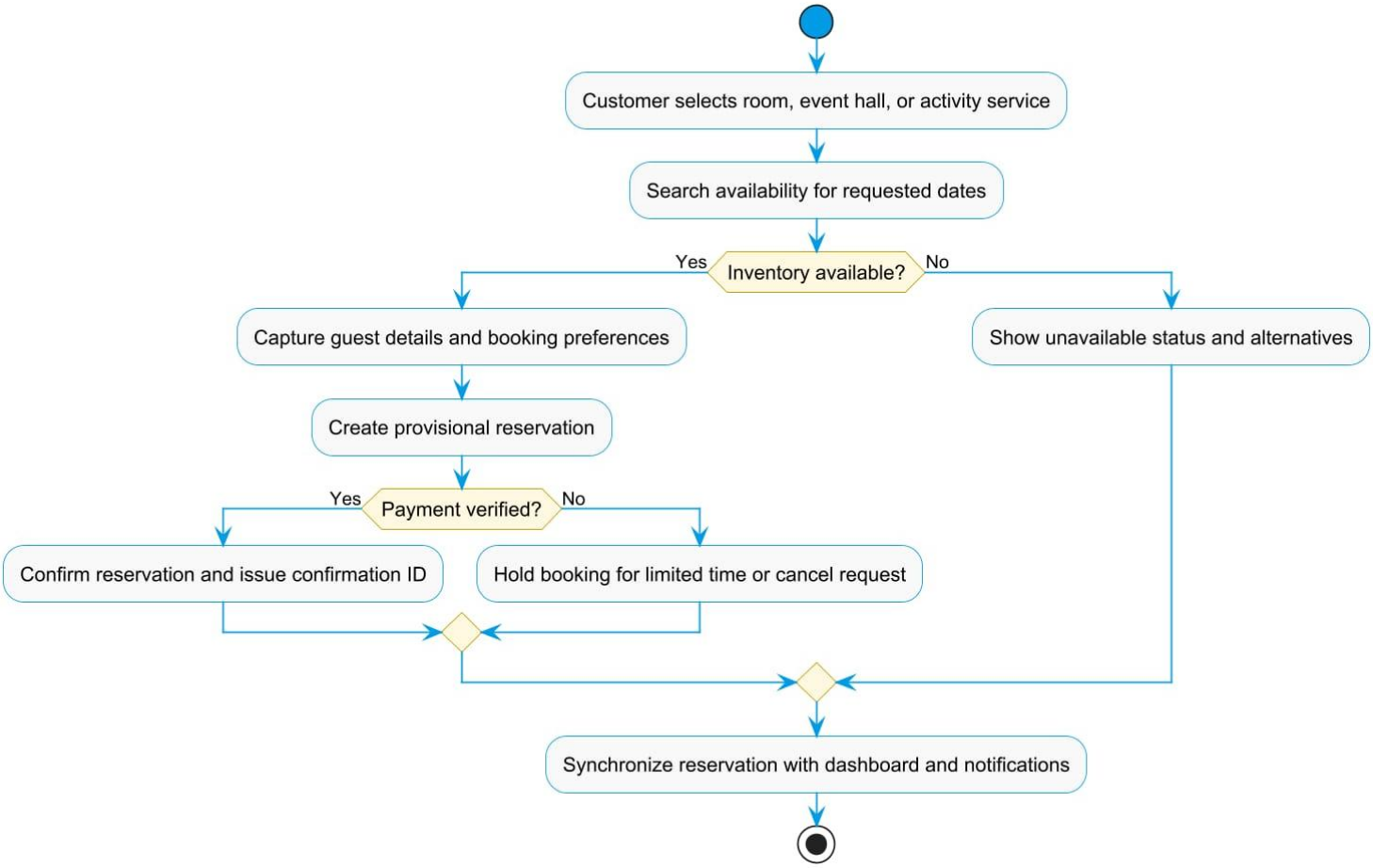
10. Event Hall & Package Management Module

Manages event halls, special packages, pricing, and reservations for events such as weddings, meetings, and conferences.



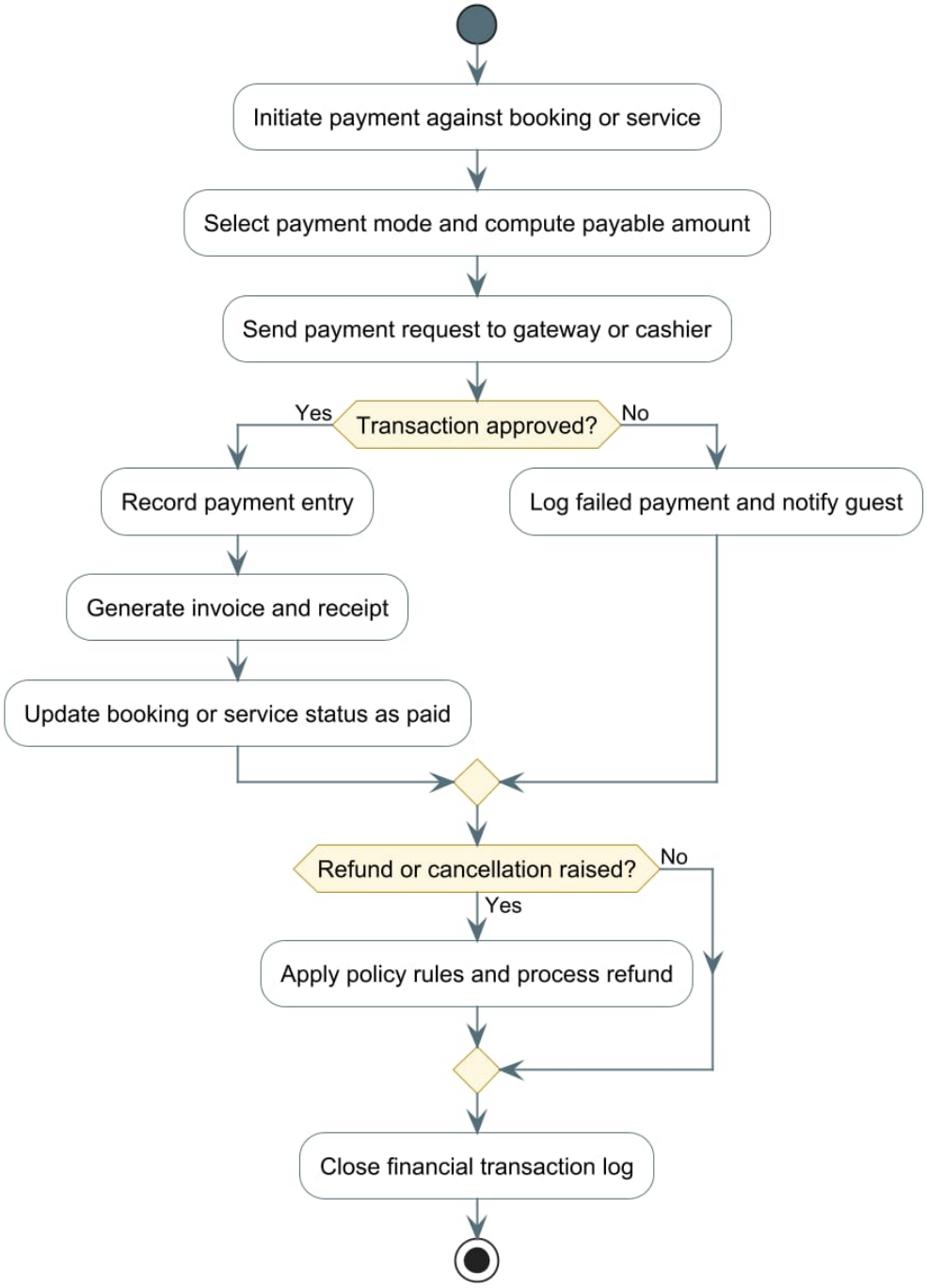
11.Booking & Reservation Management Module

Handles room, activity, and event hall reservations while managing availability and booking records.



12. Payment Management Module

Processes payments securely, maintains transaction records, and supports multiple payment methods.



Class Diagram

